

Functional Characterization of the Prejunctional Receptors Mediating the Inhibition by Ergotamine of the Rat Perivascular Sensory Peptidergic Drive

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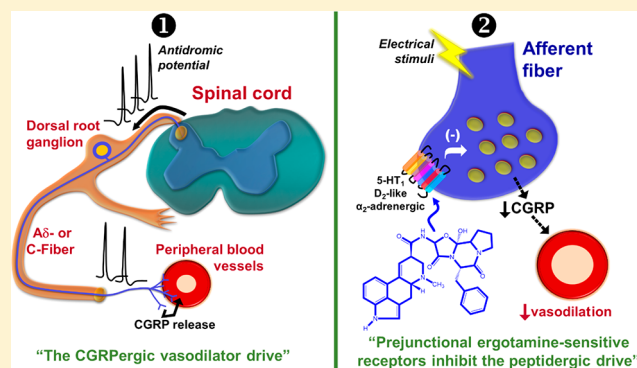
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Supporting Information

ABSTRACT: Calcitonin gene-related peptide (α -CGRP) released from perivascular sensory nerves induces decreases in diastolic blood pressure (DBP). Experimentally, this can be shown by spinal thoracic (T₉–T₁₂) electrical stimulation of these afferent fibers. Because ergotamine inhibits these neurogenic vascular responses and displays affinity for monoaminergic receptors that inhibit neurotransmitter release, we investigated whether this ergotamine-induced inhibition results from activation of serotonin 5-HT_{1B/1D}, dopamine D₂-like, and α_2 -adrenergic receptors. Wistar rats were pithed and, under autonomic ganglion blockade, received intravenous infusions of methoxamine followed by ergotamine (0.1–3.1 $\mu\text{g kg}^{-1} \text{ min}^{-1}$). Thoracic T₉–T₁₂ electrical stimulation or an intravenous bolus of α -CGRP resulted in decreases in DBP. Ergotamine inhibited the electrically induced, but not α -CGRP-induced, responses. The vasodilator sensory inhibition by 3.1 μg of ergotamine $\text{kg}^{-1} \text{ min}^{-1}$ was resistant to simultaneous blockade of 5-HT_{1B/1D}, D₂-like, and α_2 -adrenergic receptors upon addition of antagonists GR127935, haloperidol, and rauwolscine. Moreover, the inhibition by 0.31 μg of ergotamine $\text{kg}^{-1} \text{ min}^{-1}$ was unaltered by GR127935 and haloperidol, partly blocked by GR127935 and rauwolscine or rauwolscine and haloperidol, and abolished by GR127935, haloperidol, and rauwolscine. These findings imply that prejunctional 5-HT_{1B/1D}, D₂-like, and α_2 -adrenergic receptors mediate the sensory inhibition induced by 0.31 μg of ergotamine $\text{kg}^{-1} \text{ min}^{-1}$, whereas larger doses may involve other receptors. Thus, ergotamine's ability to inhibit the perivascular sensory peptidergic drive may result in facilitation of its systemic vasoconstrictor properties.

KEYWORDS: Blood pressure, CGRP, 5-HT_{1B}, D₂-like, α_2 -adrenoceptors, sensory neurons



1. INTRODUCTION

Ergotamine is a classical antimigraine agent with a complex pharmacology activating serotonergic, noradrenergic, and dopaminergic receptors.¹ Although the antimigraine therapeutic use of ergotamine has decreased due to its unspecific effects and the current availability of more selective drugs,² this ergot derivative is still used in some patients affected by sporadic migraine attacks provided that it is given as an acute treatment.^{1,2} Regardless, ergotamine may cause several side effects, including (among others) multivalvular heart disease³ and ergotism.⁴ Moreover, some basic pharmacological

questions can be answered by using ergotamine as an experimental tool.^{5–7}

Seminal studies have shown that ergotamine can induce a very potent peripheral vasoconstriction, and initially, vasoconstriction of cranial blood vessels was considered to completely explain its antimigraine properties.⁸ However, it

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has been suggested that this ergot also inhibits the trigeminal release (at peripheral and central levels) of calcitonin gene-related peptide (α -CGRP). CGRP is a neuropeptide that, in addition to participating in migraine pathophysiology,⁹ seems to decrease peripheral vascular resistance by producing a potent vasodilation.^{10–12} Indeed, this neuropeptide is localized in primary peptidergic sensory nerves¹³ surrounding resistance blood vessels.¹⁴ Although CGRP knockout mice show abnormalities in the structure of blood vessels,¹⁵ contradictory evidence has been reported with respect to the relevance of the CGRPergic system modulating blood pressure (for references, see ref 16), probably due to the experimental model used. Nevertheless, current evidence supports the notion that CGRP is relevant to counterbalance hypertensive states.^{10,12,14–16}

Experimentally, rat spinal thoracic (T₉–T₁₂) electrical stimulation results in decreases in diastolic blood pressure (DBP) that involve release of α -CGRP from primary afferent fibers,¹⁷ as these responses are selectively blocked by peptidergic (CGRP_{8–37})¹⁷ or nonpeptidergic (olcegepant)¹⁸ CGRP receptor antagonists. Furthermore, the function of sensory CGRPergic peripheral nerves (i) can be activated by several endogenous substances, including anandamide,¹⁹ prostaglandins,²⁰ bradykinin,²⁰ or acetylcholine,²¹ and (ii) can be inhibited prejunctionally by opioid peptides²² and activation of α_2 -adrenoceptors,²³ 5-HT_{1B} receptors,²⁴ 5-HT_{1F} receptors,²⁵ and H₃ receptors.²⁶

More recently, we have reported that dihydroergotamine (another ergot alkaloid used in the acute treatment of migraine) inhibits the electrically stimulated rat perivascular sensory peptidergic drive that produces decreases in DBP by activation of prejunctional α_2 -adrenergic and 5-HT_{1B/1D} receptors.²⁷ Because ergotamine can also induce a dose-dependent inhibition of these electrically induced decreases in DBP without affecting those elicited by an intravenous (iv) bolus of α -CGRP,²⁸ a prejunctional vascular sensory inhibition was suggested, although the receptors involved were not further investigated.²⁸ Hence, we have investigated the prejunctional receptors involved in ergotamine's inhibitory action on the perivascular sensory peptidergic drive producing CGRPergic decreases in DBP.

Although ergotamine displays a wide range of affinities for serotonergic, noradrenergic, and dopaminergic receptors,^{29–34} this ergot displays particularly high affinities for α_2 -adrenergic, serotonin 5-HT_{1A/1B/1D}, and dopamine D₂-like receptors (see Table 1), which are coupled to a transductional system (G_{i/o} proteins) associated with inhibition of neurotransmitter release.^{35,36} Thus, we used antagonists at 5-HT_{1B/1D} [GR127935 (GR)], D₂-like [haloperidol (HALO)], and α_2 -adrenergic [rauwolscine (RAUW)] receptors, keeping in mind the fact that prejunctional inhibition of the rat perivascular sensory peptidergic drive producing CGRPergic decreases in DBP is unrelated to activation of 5-HT_{1A/1D} receptors.²⁵

2. RESULTS AND DISCUSSION

2.1. General. This study involves the pharmacological characterization of the receptors underlying the inhibition by ergotamine of the perivascular sensory peptidergic drive. Our findings show that ergotamine displays a complex pharmacological profile in its ability to induce vascular sensory inhibition; i.e., the larger the ergotamine dose, the higher its pharmacological profile complexity (implying, at larger doses, the possible involvement of other receptors unrelated to the ones investigated in this study). Indeed, the inhibition induced

Table 1. Pharmacological Profile (pK_i values) of Ergotamine and the Antagonists Used in This Study^a

receptor	ergotamine	rauwolscine	GR127935	haloperidol
α_2	(?)8.2 ^b	nd	<6.0 ^f	5.8 ^j
α_{2A}	nd	8.9 ^d	nd	(r)4.7 ^k
α_{2B}	nd	8.9 ^d	nd	nd
α_{2C}	nd	9.3 ^d	nd	nd
D ₂ -like	(?)8.5 ^b	nd	nd	nd
D ₂	nd	nd	nd	(r)9.4 ^k
D ₃	nd	nd	nd	(r)8.5 ^k
D ₄	nd	nd	nd	(r)8.8 ^k
5-HT _{1A}	(p)8.4 ^c	nd	7.2 ^g	(r)5.7 ^k
5-HT _{1B}	(r)8.7 ^c	6.5 ^e	(r)8.8 ^h	nd
5-HT _{1D}	(p)8.4 ^c	7.8 ^e	8.6 ⁱ	nd
5-HT _{1B/1D}	7.6 ^c	nd	nd	nd
5-HT _{1E}	6.2 ^g	5.5 ^e	5.4 ⁱ	nd
5-HT _{1F}	6.8 ^g	nd	6.4 ⁱ	nd

^aThe binding affinity constants refer to cloned human receptors, except when stated otherwise. All values are pK_i values except for those in bold (pK_D). Abbreviations: nd, not determined; (p), porcine receptor; (r), rodent receptor; (?), species not specified. 5-HT_{1B/1D} refers to the, previously named, 5-HT_{1D} receptor binding site in the calf caudate membrane, at which time the experimental conditions allowed the inclusion of the 5-HT_{1B} (5-HT_{1Dp}) and 5-HT_{1D} (5-HT_{1Da}) receptor sites. ^bData from ref 34. ^cData from ref 31. ^dData from ref 59. ^eData from ref 60. ^fData from ref 61. ^gData from ref 29. ^hData from ref 62. ⁱData from ref 63. ^jData from ref 64. ^kData from ref 65.

by 0.31 μ g of ergotamine kg⁻¹ min⁻¹ may involve simultaneous activation of serotonin 5-HT_{1B} (blocked by GR, as 5-HT_{1A/1D} receptors have previously been excluded²⁵), dopamine D₂-like (blocked by HALO), and α_2 -adrenergic (blocked by RAUW) receptors, as only the GR/HALO/RAUW mixture abolished ergotamine's response (Figure 5H). In contrast, the inhibition induced by a larger infusion dose (i.e., 3.1 μ g kg⁻¹ min⁻¹) of ergotamine apparently involves other (perhaps novel) receptors that, being resistant to blockade by the same GR/HALO/RAUW mixture (Figure 4E), are unrelated to 5-HT_{1B/1D}, D₂-like, and α_2 -adrenergic receptors.

Admittedly, this study did not identify the precise 5-HT₁ receptor subtypes mediating the inhibition by ergotamine. However, it has previously been shown in dorsal root ganglion (DRG) neurons of rats that (i) functional 5-HT_{1B} and 5-HT_{1F} receptors are expressed^{37,38} and (ii) there is a lack of mRNA expression of 5-HT_{1A} or 5-HT_{1D} receptors.³⁸ In agreement with these studies, functional pharmacological experiments in pithed rats regarding the prejunctional serotonergic mechanisms modulating the perivascular sensory peptidergic drive have excluded the participation of 5-HT_{1A} and 5-HT_{1D} receptors and showed that only 5-HT_{1B} or 5-HT_{1F} receptors are involved.^{24,25} Thus, considering that GR displays a preferential affinity for 5-HT_{1B/1D} receptors, coupled to ergotamine's preferential affinity for 5-HT_{1B} (pK_i = 8.7) rather than 5-HT_{1F} (pK_i = 6.8) receptors (see Table 1), it is reasonable to suggest a main role for the 5-HT_{1B} receptor.

2.2. Effects Produced by the Different Compounds on Blood Pressure and Heart Rate. Baseline values of DBP and heart rate in the 120 pithed rats were 55 $\pm$ 6 mmHg and 254 $\pm$ 8 beats/min, respectively. The administration of gallamine or hexamethonium did not significantly change these values (not shown). Moreover, methoxamine and/or ergotamine infusions produced a significant increase in DBP

and heart rate (Figure 1 and Suppl. Table 1), as observed upon iv administration of α -CGRP (not shown). The vasopressor

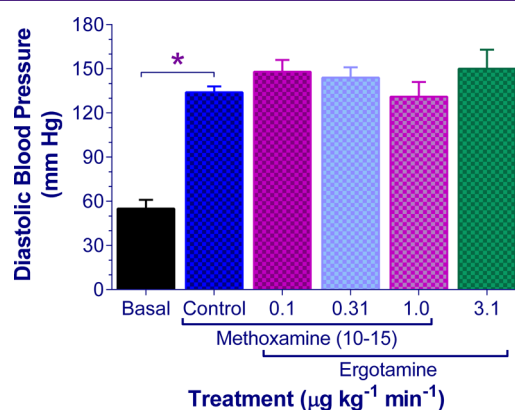


Figure 1. Increase in diastolic blood pressure before (baseline) and 20 min after the continuous iv infusions of methoxamine ($15 \mu\text{g kg}^{-1} \text{ min}^{-1}$; Control group), vehicle (0.02 mL/min; 1% PPG), and/or ergotamine (0.1 – $3.1 \mu\text{g kg}^{-1} \text{ min}^{-1}$) in the animals receiving spinal thoracic (T_9 – T_{12}) electrical stimulation. * $P < 0.05$ vs control.

response to methoxamine in rats has been attributed to activation of vascular α_1 -adrenoceptors,^{39,40} whereas that to ergotamine involves activation of vascular α_2 -adrenoceptors.⁴¹ In addition, although ergotamine has been considered a potent vasoconstrictor agent per se, under our experimental conditions (except for $3.1 \mu\text{g}$ of ergotamine $\text{kg}^{-1} \text{ min}^{-1}$), we needed to infuse methoxamine (10 – $15 \mu\text{g kg}^{-1} \text{ min}^{-1}$) simultaneously with 0.1 – $1 \mu\text{g}$ of ergotamine $\text{kg}^{-1} \text{ min}^{-1}$ because the latter doses of ergotamine, when given alone, produced only small and inconsistent increases in DBP (data

not shown). These experimental conditions with the infusions of methoxamine and/or ergotamine allowed us to produce a sustained increase in DBP (Figures 1 and 2 and Suppl. Table 1) for producing consistent CGRPergic decreases in DBP.

Furthermore, the tachycardic response produced by the infusions of methoxamine and/or ergotamine (Suppl. Table 1) may be attributed to cardiac activation of α_1 -adrenoceptors,⁴² as suggested in similar studies.^{23,28} Moreover, a significant decrease in DBP was produced immediately after iv administration of RAUW, HALO, or their combinations (except GR alone), but this effect returned to baseline values after 10 min (Suppl. Tables 2 and 3).

It should be noted that iv saline, RAUW, GR, or HALO given alone, as well as the RAUW/GR, RAUW/HALO, or GR/HALO mixtures, did not have significant effects ($P > 0.05$) on the rat perivascular sensory peptidergic drive that produces decreases in DBP.²⁷ Similar effects ($P > 0.05$) were observed with the iv GR/HALO/RAUW mixture (not shown).

2.3. Effects of Ergotamine on the CGRPergic Decreases in DBP. As shown in Figure 2, electrical stimulation of the sensory peptidergic drive (Figure 2A) or exogenous α -CGRP (Figure 2B) produced consistent decreases in DBP; as reported previously,²⁸ these responses were not accompanied by noticeable effects on heart rate. Moreover, during the infusions of ergotamine, only the electrically induced (but not the α -CGRP-induced) decreases in DBP (Figure 2C) were inhibited [as compared to control experiments (see Figure 2A)]. Accordingly, Figure 3 shows that electrical stimulation of the sensory peptidergic drive produced decreases in DBP that (i) were not affected by vehicle [1% PPG, 0.02 mL/min (Figure 3A)] and (ii) were attenuated by ergotamine, particularly at doses of 0.31 – $3.1 \mu\text{g kg}^{-1} \text{ min}^{-1}$ (Figure 3B). Furthermore, panels C and D of

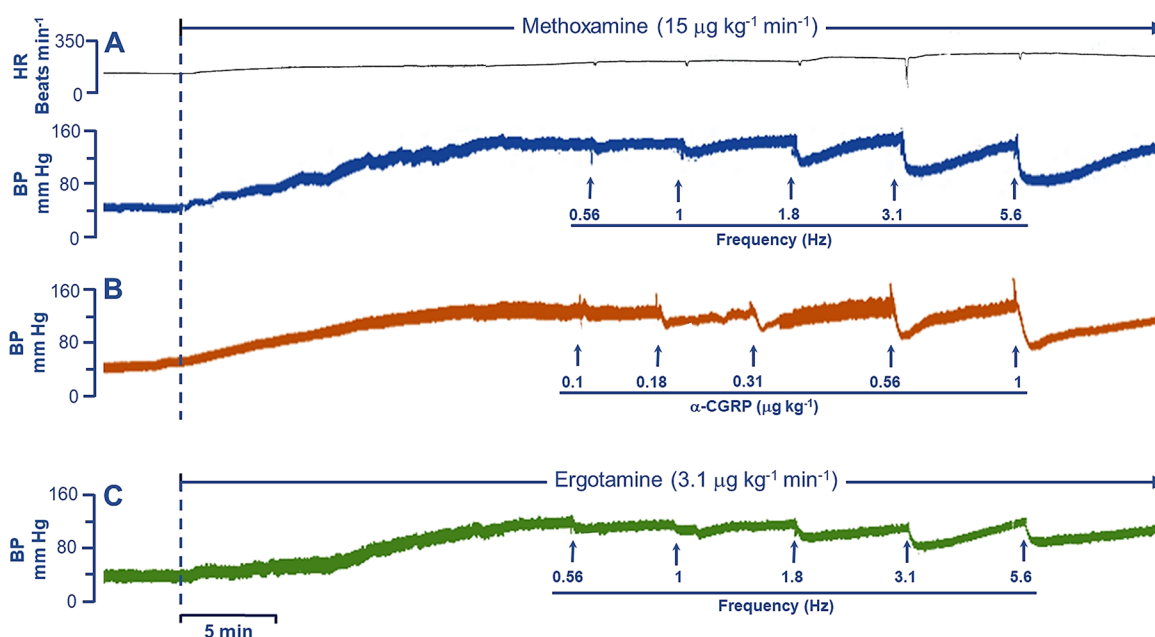


Figure 2. Experimental tracings illustrating the vasodepressor responses induced by electrical stimulation of the perivascular sensory drive or iv bolus injections of exogenous α -CGRP. (A and B) Effects of electrical stimulation and α -CGRP, respectively, on diastolic blood pressure during the continuous infusion of methoxamine (control). In both cases, the decreases in blood pressure were accompanied by small and inconsistent changes in heart rate. (C) Effect of ergotamine ($3.1 \mu\text{g kg}^{-1} \text{ min}^{-1}$) on the vasodepressor responses induced by electrical stimulation. In contrast with the methoxamine control curve, ergotamine inhibits the vasodepressor responses induced by electrical stimulation but not by α -CGRP (not shown); similar results were obtained during infusions of $0.31 \mu\text{g}$ of ergotamine $\text{kg}^{-1} \text{ min}^{-1}$. This finding suggests that the inhibition of the vasodepressor responses by ergotamine is prejunctional in nature. BP, blood pressure; HR, heart rate.

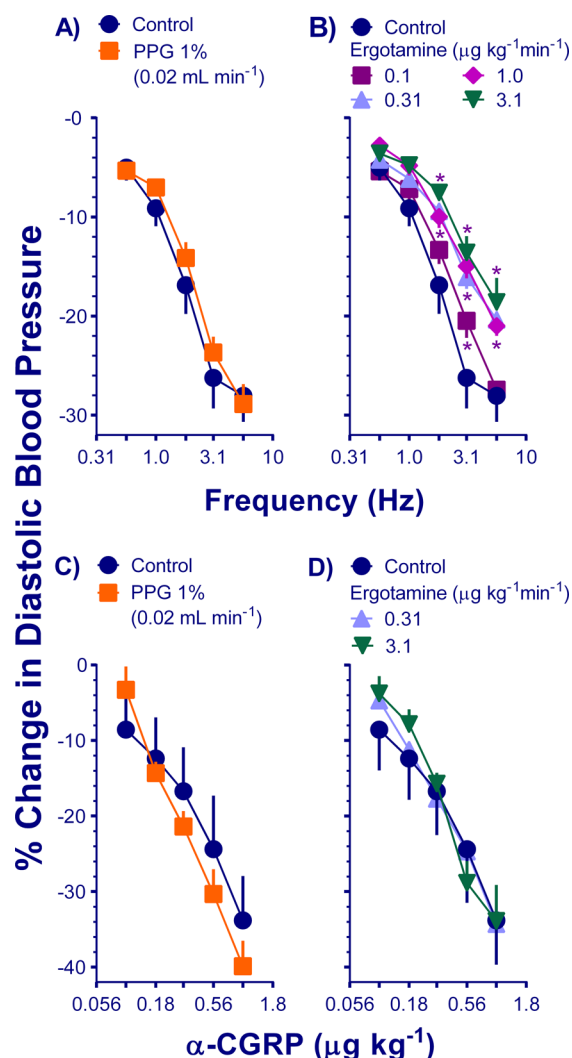


Figure 3. Vasodepressor responses induced by (A and B) electrical stimulation or (C and D) iv bolus injections of exogenous α -CGRP elicited during continuous iv infusions of vehicle (1% PPG, 0.02 mL/min) or ergotamine (0.31–3.1 $\mu\text{g kg}^{-1} \text{min}^{-1}$) ($n = 5$ each dose). * $P < 0.05$ vs control.

Figure 3 show that exogenous α -CGRP also elicited decreases in DBP; these responses, which were not affected by vehicle [1% PPG, 0.02 mL/min (Figure 3C)], remained unchanged by ergotamine [0.31 and 3.1 $\mu\text{g kg}^{-1} \text{min}^{-1}$ (Figure 3D)]. Taken together, these results in pithed rats allow us (i) to rule out a possible physiological antagonism by the direct vascular effects of ergotamine and (ii) to suggest a vascular sensory inhibition mechanism, as previously described for clonidine,²³ sumatriptan,²⁵ immepip,²⁶ dihydroergotamine,²⁷ and quipir-ole.⁴³

In light of the fact that 0.31 and 3.1 μg of ergotamine $\text{kg}^{-1} \text{min}^{-1}$, which were capable of inhibiting (to the same degree) the electrically induced decreases in DBP (Figure 3B), failed to significantly affect those by exogenous α -CGRP, these infusion doses of ergotamine were selected for further analysis with antagonists.

2.4. Contribution of Prejunctional 5-HT_{1B}, D₂-like, and α_2 -Adrenergic Receptors to the Inhibition by Ergotamine of the Rat Perivascular Sensory Peptidergic Drive. The possible correlation of the sensory inhibition by ergotamine with 5-HT_{1B}, D₂-like, and α_2 -adrenergic receptors

was evaluated by using effective blocking doses of the antagonists GR (5-HT_{1B/1D}), HALO (D₂-like), and RAUW (α_2 -adrenergic) (Table 1), as reported previously.^{23–25,27,43–47}

Hence, Figures 4 and 5 show the effects of these antagonists on ergotamine's sensory inhibition. In this respect, the sensory inhibition by 3.1 μg of ergotamine $\text{kg}^{-1} \text{min}^{-1}$ remained unaltered in animals treated with vehicle (saline) (Figure 4A), RAUW (Figure 4B), GR (Figure 4C), or HALO (Figure 4D) given alone or after the mixture of the three antagonists (Figure 4E). These results may suggest, at first glance, that α_2 -adrenergic, 5-HT_{1B/1D}, and D₂-like receptors are not involved and would apparently imply the possible disclosure of another inhibitory (maybe novel) mechanism (see below). To ascertain the relevance of the ergotamine dose employed (i.e., 3.1 $\mu\text{g kg}^{-1} \text{min}^{-1}$), we decided to further analyze the inhibition by a smaller dose (i.e., 0.31 $\mu\text{g kg}^{-1} \text{min}^{-1}$) of ergotamine. Hence, Figure 5 shows that the inhibition by 0.31 μg of ergotamine $\text{kg}^{-1} \text{min}^{-1}$ was (i) unaffected by vehicle (Figure 5A), RAUW (Figure 5B), GR (Figure 5C), HALO (Figure 5D), or GR and HALO (Figure 5F); (ii) partially blocked by RAUW and GR (Figure 5E) or RAUW and HALO (Figure 5G); and (iii) abolished by the GR/HALO/RAUW mixture (Figure 5H). At this point, we must keep in mind that the doses of these antagonists were, by themselves, devoid of any effect on the rat perivascular sensory peptidergic drive, as previously reported.^{23–25,27,43}

These findings imply that the sensory inhibition by 0.31 μg of ergotamine $\text{kg}^{-1} \text{min}^{-1}$ may involve co-activation of 5-HT_{1B/1D}, D₂-like, and α_2 -adrenergic receptors. Admittedly, as mentioned in section 2.1, ergotamine has a high affinity for 5-HT_{1A} and 5-HT_{1D} receptors (Table 1), but these receptors have previously been excluded from the inhibition of the perivascular sensory peptidergic drive.^{25,37,38} Moreover, the fact that the inhibition by 0.31 μg of ergotamine $\text{kg}^{-1} \text{min}^{-1}$ was not affected by GR, HALO, or RAUW given alone highlights the relevance of 5-HT_{1B}, D₂-like, and α_2 -adrenergic receptors in the modulation of the CGRPergic neurotransmission at the neurovascular level. Accordingly, our results imply that this dose of ergotamine may be equipotently and simultaneously stimulating 5-HT_{1B}, D₂-like, and α_2 -adrenergic receptors. Following this line of reasoning, the blockade of one of these receptors would be masked by the sensory inhibition produced by the unblocked receptors. Apart from the role of 5-HT_{1B} and D₂-like receptors, the function of α_2 -adrenergic receptors modulating perivascular CGRPergic transmission has been demonstrated *in vitro*⁴⁸ and in pithed rats.²³ Indeed, in the latter experimental model, the specific role of the α_{2A} and α_{2C} , but not of the α_{2B} , subtypes has been demonstrated.²³ Thus, although the affinity of ergotamine for these subtypes remains unknown (Table 1), the α_{2A} and/or α_{2C} subtypes may be involved, because RAUW has similar affinities for these subtypes (Table 1).

Moreover, the presence of D₂-like receptors in DRG-projecting sensory neurons has been, if at all, exiguously investigated. Only one study has established the expression of D₂-like receptors in primary sensory neurons,⁴⁹ and another one from our group has pharmacologically shown the role of these receptors inhibiting the rat perivascular sensory peptidergic drive.⁴³ Clearly, additional experiments investigating the specific D₂-like receptor subtypes (D₂, D₃, or D₄)⁵⁰ involved in the inhibition of the perivascular sensory peptidergic drive are required.

2.5. Final Considerations of the Possible Mechanisms Involved in the Inhibition by 3.1 μg of Ergotamine kg^{-1}

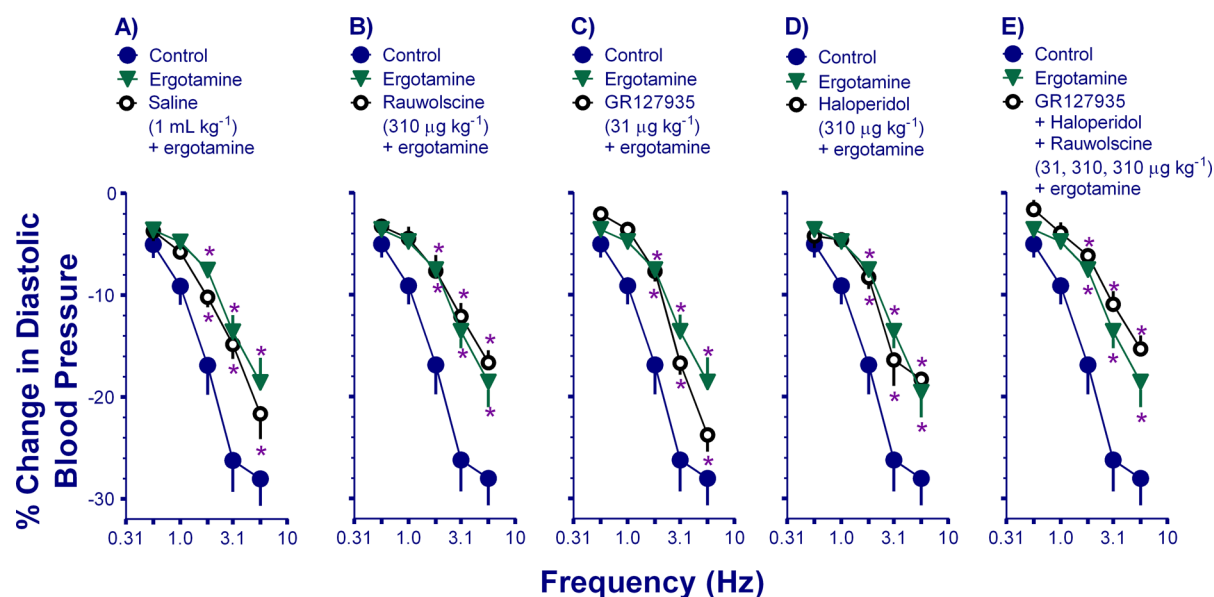


Figure 4. Effect of iv bolus injections of (A) saline (1 mL kg^{-1}) or the antagonists (B) rauwolschine ($310 \mu\text{g kg}^{-1}$), (C) GR127935 ($31 \mu\text{g kg}^{-1}$), (D) haloperidol ($310 \mu\text{g kg}^{-1}$), and (E) GR127935, haloperidol, and rauwolschine (same doses) on the inhibition of the perivascular sensory peptidergic drive induced by $3.1 \mu\text{g}$ of ergotamine $\text{kg}^{-1} \text{ min}^{-1}$. The control group represents that of animals receiving a continuous iv infusion of methoxamine ($15 \mu\text{g kg}^{-1} \text{ min}^{-1}$), which is shown for comparison. $*P < 0.05$ vs control.

min^{-1} of the Perivascular Sensory Peptidergic Drive.

Admittedly, whether additional receptors are involved in the inhibition by $3.1 \mu\text{g}$ of ergotamine $\text{kg}^{-1} \text{ min}^{-1}$ remains unknown. Hence, it seems reasonable to assume that, besides 5-HT_{1B} , D_2 -like, and α_2 -adrenergic receptors, additional receptors producing sensory inhibition that overshadow the blockade of the receptors mentioned above may play a role. In this respect, it should be kept in mind that ergotamine also displays a high affinity for recombinant 5-ht_{SA} ($pK_i = 8.4$), 5-ht_{SB} ($pK_d = 8.5$),⁵¹ and even unknown (orphan, novel, unclassified) receptors. Indeed, $5\text{-ht}_{SA/SB}$ receptors inhibit the cardiac sympathetic drive.⁵² In this context, although methiothepin and methysergide may be used as antagonists for 5-ht_{SA} (pK_i values of 7.0 and 7.2, respectively) and 5-ht_{SB} (pK_i values of 7.8 and 6.9, respectively) receptors,³² we can foresee some experimental limitations. The major concern is that methiothepin and methysergide display affinity for 5-HT_{2B} (pK_i values of 9.23 and 9.19, respectively)⁵³ and 5-HT_7 (pK_i values of 9.36 and 7.58, respectively)⁵⁴ receptors, and activation of vascular 5-HT_7 or endothelial 5-HT_{2B} receptors would result in vasodilation and/or decreases in DBP per se.^{55,56} These decreases in DBP, under our experimental conditions, would represent an impediment to analyzing the mechanisms involved in the inhibition induced by $3.1 \mu\text{g}$ of ergotamine $\text{kg}^{-1} \text{ min}^{-1}$. Finally, on balance, it could be argued that using larger doses of GR, HALO, and RAUW (alone and in mixtures as described above) would block the sensory inhibition by $3.1 \mu\text{g}$ of ergotamine $\text{kg}^{-1} \text{ min}^{-1}$. However, the doses of GR, HALO, and RAUW mentioned above produce a complete blockade of the $5\text{-HT}_{1B/1D}$,^{19,20} D_2 -like,³⁷ and α_2 -adrenergic¹⁸ receptors, respectively, that inhibit the rat perivascular sensory peptidergic drive.

3. CONCLUSION

This study suggests that, at $0.31 \mu\text{g kg}^{-1} \text{ min}^{-1}$, ergotamine inhibits the rat perivascular sensory peptidergic drive by stimulation of $5\text{-HT}_{1B/1D}$, D_2 -like, and α_2 -adrenergic

receptors on sensory CGRPergic nerves. Interestingly, larger doses of this ergot ($3.1 \mu\text{g kg}^{-1} \text{ min}^{-1}$) seem to interact with additional (perhaps novel) receptors and/or mechanisms producing sensory inhibition, which may mask the blockade of the receptors mentioned above. These findings, which distinguish ergotamine's complex pharmacological properties from those recently described for dihydroergotamine,²⁷ may help explain its cardiovascular side effects when used in migraine treatment.

4. METHODS

4.1. Animals. All animal procedures and the protocols of this investigation were approved by our Institutional Ethics Committee on the use of animals in scientific experiments (CICUAL-Cinvestav; Permission Protocol 0139-15) and followed the regulations established by the Mexican Official Norm (NOM-062-ZOO-1999), according to the Guide for the Care and Use of Laboratory Animals in the United States. Male Wistar normotensive rats (300–350 g) were maintained on a 12 h/12 h light/dark cycle (with light beginning at 07:00 a.m.) and housed in a special room at a constant temperature ($22 \pm 2^\circ\text{C}$) and humidity (50%), with food and water freely available in their home cages.

4.2. General Methods. Experiments were carried out in a total of 120 rats. After anesthesia with ether and cannulation of the trachea, the rats were pithed by inserting a stainless steel rod through the orbit and foramen magnum into the vertebral foramen.⁵⁷ Then, the animals were artificially ventilated with room air using a model 7025 Ugo Basile pump (56 strokes/min, stroke volume of 20 mL kg^{-1}), as previously described.⁵⁸ After bilateral vagotomy, catheters were placed in (i) the left and right femoral and jugular veins, for the continuous infusions of hexamethonium and agonists (methoxamine and/or ergotamine) and iv administration of antagonists, respectively, and (ii) the left carotid artery, connected to a Grass pressure transducer (P23XL), for the recording of arterial blood pressure. The heart rate was measured with a tachograph (7P4, Grass Instrument Co., Quincy, MA), triggered from the blood pressure signal. Both the blood pressure and the heart rate were recorded simultaneously by a model 7 Grass polygraph (Grass Instrument Co.). At this point, the 120 rats were initially divided into two main sets (see Figure 6), so that the effects produced by the continuous infusions of methoxamine and/or

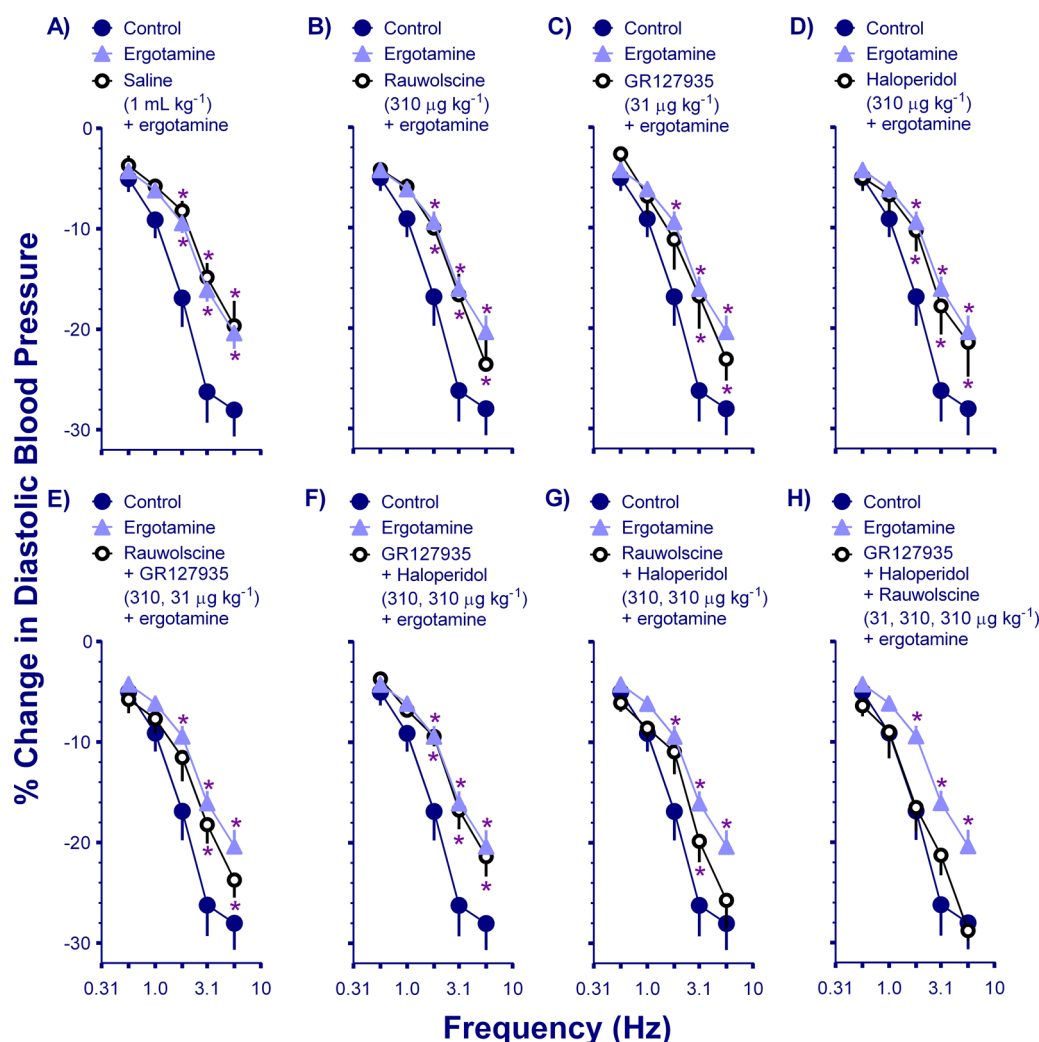


Figure 5. Effect of iv bolus injections of (A) saline (1 mL kg^{-1}) or the antagonists (B) rauwolscline ($310 \mu\text{g kg}^{-1}$), (C) GR127935 ($31 \mu\text{g kg}^{-1}$), (D) haloperidol ($310 \mu\text{g kg}^{-1}$), (E) rauwolscline and GR127935, (F) GR127935 and haloperidol, (G) rauwolscline and haloperidol, or (H) GR127935, haloperidol, and rauwolscline on the inhibition of the perivascular sensory peptidergic drive induced by $0.31 \mu\text{g}$ of ergotamine $\text{kg}^{-1} \text{ min}^{-1}$. The control group represents the group of animals receiving a continuous iv infusion of methoxamine ($15 \mu\text{g kg}^{-1} \text{ min}^{-1}$), which is shown for comparison. $*P < 0.05$ vs control.

ergotamine under different treatments could be investigated with respect to the decreases in DBP induced by (i) electrical stimulation of the perivascular (vasodepressor) sensory drive (set 1; $n = 95$) or (ii) iv bolus injections of exogenous α -CGRP (set 2; $n = 25$). The decreases in DBP elicited by electrical stimulation or exogenous α -CGRP were completed in ~ 50 min, and each response was elicited under unaltered values of resting blood pressure. The electrical stimuli (0.56, 1, 1.8, 3.1, and 5.6 Hz) as well as the dosing with α -CGRP (0.1, 0.18, 0.31, 0.56, and $1 \mu\text{g kg}^{-1}$) were given using a sequential schedule at 5–10 min intervals (see below), as previously reported.¹² The body temperature of each pithed rat was maintained at 37°C by a lamp and monitored with a rectal thermometer.

4.3. Experimental Protocols. After the animals ($n = 120$) had been in a stable hemodynamic condition for at least 10 min, baseline values of DBP (a more accurate indicator of peripheral vascular resistance) and heart rate were determined.

4.3.1. Electrical Stimulation of the Perivascular Sensory Drive. In the first set of rats [$n = 95$ (see Figure 6)], the pithing rod was replaced by an electrode enameled except for a 1.5 cm length 9 cm from the tip, so that the uncovered segment was situated at the thoracic T_9 – T_{12} segments of the spinal cord, and an indifferent electrode was placed dorsally.^{23,57} Before electrical stimulation, the animals received (iv) (i) a bolus injection of gallamine (25 mg kg^{-1}), to avoid the electrically induced muscular twitching, and (ii) 10 min

later a continuous infusion of hexamethonium ($2 \text{ mg kg}^{-1} \text{ min}^{-1}$), to block the vasopressor responses produced by stimulation of the preganglionic sympathetic vasopressor drive. Ten minutes later, this set of rats was divided into three groups ($n = 30, 25$, and 40).

The first group ($n = 30$) was subdivided into three subgroups comprising continuous iv infusions of (i) methoxamine ($15 \mu\text{g kg}^{-1} \text{ min}^{-1}$; $n = 5$, control group), (ii) ergotamine ($3.1 \mu\text{g kg}^{-1} \text{ min}^{-1}$; $n = 5$), and (iii) methoxamine (10 – $15 \mu\text{g kg}^{-1} \text{ min}^{-1}$) with vehicle [1% propylene glycol (PPG), 0.02 mL/min ; $n = 5$] or ergotamine (0.1, 0.31, or $1 \mu\text{g kg}^{-1} \text{ min}^{-1}$; $n = 5$ for each infusion dose). Twenty minutes after the infusion of the respective treatment, the DBP and heart rate were determined again, and then, the perivascular sensory drive was electrically stimulated during the treatments described above to elicit decreases in DBP by applying 10 s trains of monophasic, rectangular pulses (2 ms, 50 V), at increasing frequencies of stimulation (0.56, 1, 1.8, 3.1, and 5.6 Hz), as previously described.²³ When the DBP had returned to baseline levels, the next frequency was applied. This procedure was systematically performed until the stimulus–response curve had been completed.

The second group ($n = 25$) received a continuous iv infusion of ergotamine ($3.1 \mu\text{g kg}^{-1} \text{ min}^{-1}$). Ten minutes later, this group was subdivided into five subgroups ($n = 5$ each) comprising iv bolus injections of (i) saline (1 mL kg^{-1}), (ii) RAUW ($310 \mu\text{g kg}^{-1}$), (iii) GR ($31 \mu\text{g kg}^{-1}$), (iv) HALO ($310 \mu\text{g kg}^{-1}$), and (v) a mixture of GR,

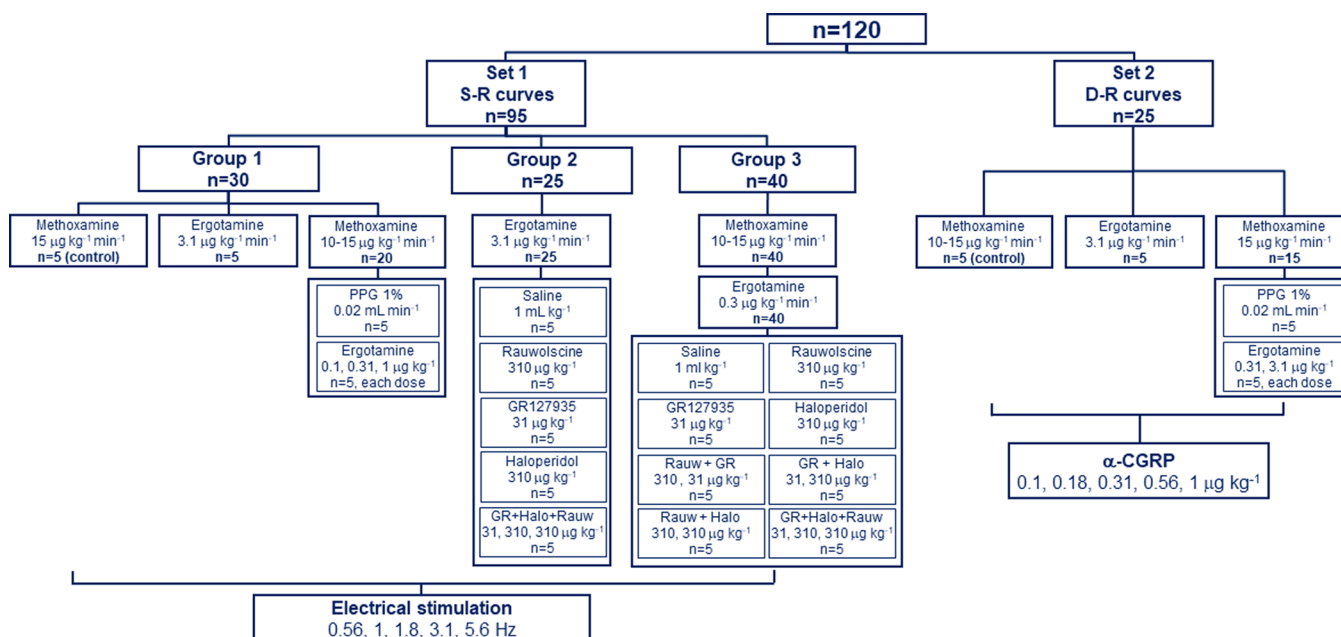


Figure 6. Experimental protocols showing the number of animals used in the two main sets and the different groups and subgroups used in this study. Methoxamine and/or ergotamine was infused during and until the end of the experiments. Abbreviations: S-R, stimulus–response curve; D-R, dose–response curve to exogenous α -CGRP; Rauw, rauwolscine; GR, GR127935; and Halo, haloperidol.

HALO, and RAUW (31, 310, and 310 $\mu\text{g kg}^{-1}$, respectively) as previously validated.²⁷ After 10 min, a stimulus–response curve was constructed as described above.

The third group ($n = 40$) received a continuous iv infusion of methoxamine (10–15 $\mu\text{g kg}^{-1} \text{min}^{-1}$) followed, 10 min later, by an infusion of ergotamine (0.31 $\mu\text{g kg}^{-1} \text{min}^{-1}$). After 10 min, this group was subdivided into eight subgroups ($n = 5$ each) comprising iv bolus injections of (i) saline (1 mL kg^{-1}), (ii) RAUW (310 $\mu\text{g kg}^{-1}$), (iii) GR (31 $\mu\text{g kg}^{-1}$), (iv) HALO (310 $\mu\text{g kg}^{-1}$), (v) RAUW and GR (310 and 31 $\mu\text{g kg}^{-1}$, respectively), (vi) GR and HALO (31 and 310 $\mu\text{g kg}^{-1}$, respectively), (vii) RAUW and HALO (310 $\mu\text{g kg}^{-1}$ each), and (viii) a mixture of GR, HALO, and RAUW (31, 310, and 310 $\mu\text{g kg}^{-1}$, respectively), in doses previously validated to block their respective receptors in pithed rats.²⁷ Ten minutes later, a stimulus–response curve was constructed as described above.

4.3.2. Administration of Exogenous α -CGRP. The second set of rats [$n = 25$ (see Figure 6)] was prepared as describe above, but the pithing rod was left throughout the experiments and the administration of both gallamine and hexamethonium was omitted. This set was subdivided into three subgroups comprising continuous iv infusions of (i) methoxamine (15 $\mu\text{g kg}^{-1} \text{min}^{-1}$; $n = 5$, control group), (ii) ergotamine (3.1 $\mu\text{g kg}^{-1} \text{min}^{-1}$; $n = 5$), or (iii) methoxamine (10–15 $\mu\text{g kg}^{-1} \text{min}^{-1}$) with vehicle (1% PPG, 0.02 mL/min; $n = 5$) or ergotamine (0.31 or 3.1 $\mu\text{g kg}^{-1} \text{min}^{-1}$; $n = 5$ each group). Twenty minutes after the infusion of the respective treatment had started, the values of DBP and heart rate were determined again, and then, the decreases in DBP produced by iv bolus injections of exogenous α -CGRP (0.1, 0.18, 0.31, 0.56, and 1 $\mu\text{g kg}^{-1}$) were examined.

4.3.3. Other Procedures Applying to All Protocols. The doses of hexamethonium, methoxamine, and ergotamine were continuously infused at a rate of 0.02 mL/min by a WPI model sp100i pump (WPI, Inc., Sarasota, FL). The interval between the different frequencies of stimulation or doses of α -CGRP were dependent on the duration of the resulting decreases in DBP (usually 5–10 min), as we waited until the DBP had returned to baseline values.

4.4. Compounds. Apart from the anesthetic (diethyl ether), the compounds used in this study (obtained from the sources indicated) were gallamine triethiodide, hexamethonium chloride, rat α -CGRP, methoxamine hydrochloride, and rauwolscine hydrochloride (Sigma Chemical Co., St. Louis, MO); N-[methoxy-3-(4-methyl-1-

piperazinyl)phenyl]-2'-methyl-4'-(5-methyl-1,2,4-oxadiazol-3-yl)[1,1-biphenyl]-4-carboxamide hydrochloride (GR127935) (GlaxoSmithKline, Stevenage, Hertfordshire, U.K.); and ergotamine tartrate (gift from Novartis Pharma A.G., Basel, Switzerland). All compounds were dissolved in saline, except (i) 1% (v/v) PPG to dissolve ergotamine (the resulting solution was finally diluted with saline) and (ii) 5% (w/v) ascorbic acid to dissolve haloperidol (the resulting solution was finally diluted with saline); 1% PPG had no effect per se on (i) the baseline diastolic blood pressure and heart rate (data not shown) or (ii) the decreases in DBP induced electrically or by exogenous α -CGRP (Figure 3). An ascorbic acid concentration of 5% was not evaluated in terms of these variables because haloperidol (dissolved in this vehicle) had no effect per se on the rat perivascular peptidergic drive.²⁷ Fresh solutions were prepared for each experiment. The doses of agonists refer to their respective salts, whereas those of the antagonists refer to their free base.

4.5. Statistical Analysis of Data. All data in the text, tables, and figures are presented as means $\pm$ SEM. The peak changes in DBP by electrical stimulation or exogenous α -CGRP were expressed as the percent change from baseline, as previously described in rats.^{23–25} The differences in the absolute values of DBP and heart rate within one subgroup of animals before and during the continuous infusions of methoxamine (10–15 $\mu\text{g kg}^{-1} \text{min}^{-1}$) and/or ergotamine were evaluated with a paired Student's t test. Moreover, a one-way repeated measures analysis of variance was used to compare the absolute values of DBP and heart rate obtained during the continuous infusions of methoxamine and/or ergotamine before, immediately after, and 10 min after the administration of saline or the antagonists used. Furthermore, a one-way analysis of variance was used to compare the DBP before (baseline) and during methoxamine (control) or ergotamine infusions.

Finally, the decreases in DBP induced electrically or by exogenous α -CGRP in the different subgroups of animals were compared with a two-way analysis of variance. The one- and two-way analyses of variance were followed, if applicable, by the Bonferroni post hoc test. Statistical significance was accepted at the $P < 0.05$ level.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscchemneuro.8b00611.

Suppl. Tables 1–3 listing relevant cardiovascular data (effects of the different treatments on diastolic blood pressure and heart rate) (PDF)

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

CGRP, calcitonin gene-related peptide; DBP, diastolic blood pressure; DRG, dorsal root ganglion; GR, GR127935; HALO, haloperidol; PPG, propylene glycol; RAUW, rauwolscine; SEM, standard error of the mean

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